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A Human-in-the-Loop Environment for Arabic Machine Translation

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Abstract

This report details a comprehensive project to develop a human-in-the-loop (HITL) translation environment for English–Arabic and French–Arabic language pairs. The work encompasses the curation of a large-scale, multi-domain parallel corpus; the systematic evaluation and selection of state-of-the-art multilingual translation models; and the design of a sophisticated agentic pipeline for continuous learning from human feedback. Key findings demonstrate that domain-specific fine-tuning using parameter-efficient methods (QLoRA) significantly enhances translation quality, with the NLLB-3.3B model achieving the best overall performance. The proposed system architecture integrates multiple large language models (LLMs) into a translator-centric workflow, prioritizing human judgment, consistency, and cultural appropriateness over fully automated translation. The project contributes high-quality Arabic language resources and presents a scalable framework for adaptive, human-AI collaborative translation.

1 Introduction

Translation plays a vital role in enabling communication between people, cultures, and institutions across the world. It allows knowledge to circulate beyond linguistic boundaries, supports education and research, and facilitates cooperation in fields such as science, technology, business, and governance. In today’s digital age, the demand for fast and accurate translation has increased significantly, especially with the growth of online content and international collaboration. As a result, machine translation systems have become widely used to assist in handling large volumes of multilingual text.

Although machine translation has improved significantly, it still struggles with complex languages such as Arabic. The language’s rich morphology, flexible structure, and context dependence, along with cultural and domain-specific expressions, often lead to inaccurate or unnatural translations, reducing trust in fully automated systems, especially in professional contexts.

For these reasons, human translators continue to play an essential role in ensuring translation quality. Human expertise is crucial for understanding context, preserving meaning, and producing fluent and culturally appropriate translations. However, professional translation is time-consuming and cognitively demanding, particularly when translators must work with long documents or highly technical content. Repetitive tasks and terminology consistency can significantly increase workload and reduce productivity. This creates a need for tools that support human translators rather than replace them.

Recent advances in large language models (LLMs) offer promising opportunities for translation assistance. They are effective at understanding context, generating fluent text, and adapting to different domains, providing high-quality translation suggestions and stylistic alternatives. However, relying on a single model or fully automated output remains risky, as errors, hallucinations, or biased content may occur, making human supervision essential.

This project addresses these challenges by proposing a **human-in-the-loop translation environment** for English–Arabic and French–Arabic translation. Instead of replacing the human translator, the system is designed to place the human at the center of the translation process, supported by multiple LLMs. The main idea is to combine the strengths of advanced machine translation models with human judgment, expertise, and cultural awareness. By doing so, the system aims to improve translation quality while also enhancing productivity and reducing repetitive effort.

The proposed solution integrates several key components into a single environment. It includes curated bilingual data, fine-tuned open-source LLMs, and a translator-oriented interface that allows users to compare, edit, and refine translations efficiently. Translators will be able to upload documents, view multiple translation outputs side

by side, select a reference translation, and interactively edit the text using suggestions from different models. This approach encourages informed decision-making and gives translators full control over the final output.

The importance of this project lies not only in its technical contribution but also in its practical impact. By focusing on usability, safety, and cultural relevance, the system aims to produce translations that are accurate, trustworthy, and suitable for real-world professional use. Moreover, by optimizing the solution for limited computational resources and supporting both web-based and desktop deployment, the project seeks to make advanced translation assistance accessible in a wide range of contexts.

In collaboration with the Algerian Academy for the Arabic Language, this project also contributes to the development of high-quality Arabic language resources, including parallel corpora and translation memories. These resources can support future research and applications in Arabic natural language processing.

Overall, this project proposes a balanced and human-centered approach to machine translation. By integrating LLMs into a controlled, interactive environment, it demonstrates how artificial intelligence can effectively support human translators while preserving quality, trust, and cultural sensitivity. This human–AI collaboration model has the potential to serve as a reference for similar translation systems in other language pairs and domains.

2 State of the Art

Machine translation (MT) has advanced rapidly with large language models (LLMs) built on Transformers. Unlike earlier neural MT (NMT) systems trained on parallel corpora, modern LLMs are trained on vast multilingual text and can perform translation as one capability among many. This section reviews LLM-based MT with a focus on Arabic, compares commercial and open-source approaches, and surveys human-in-the-loop (HITL) translation environments.

2.1 LLM-Based Machine Translation

General-purpose LLMs can achieve translation quality comparable to dedicated MT systems, especially for high-resource languages. LLMs excel at leveraging broader context, maintaining coherence across paragraphs or documents, and reducing inconsistencies from sentence-by-sentence translation. They can be steered via prompting (instructions, examples, style constraints) for rapid adaptation.

However, performance is uneven across languages and domains, declining in low-resource settings or for underrepresented dialects/registers. LLM outputs can be variable; small prompt changes affect translation choices, challenging terminological consistency. Consequently, recent work treats LLM translation as a strong first draft benefiting from structured guidance (terminology constraints, translation memory retrieval, human post-editing).

2.2 Arabic Machine Translation with LLMs

Arabic translation remains challenging due to linguistic and resource factors:

- **Morphological richness & ambiguity:** Words encode multiple features via prefixes/suffixes/clitics, with diacritics often omitted, increasing ambiguity.
- **Diglossia & dialectal variation:** Tension between Modern Standard Arabic (MSA) and numerous under-resourced dialects differing in vocabulary, orthography, and syntax.
- **Word order flexibility & stylistic constraints:** Flexible syntax and varying conventions across domains (news, legal, religious, marketing) require careful choices for formality and idiomaticity.
- **Data scarcity & domain mismatch:** High-quality parallel corpora are limited compared to English-centric resources, with datasets often skewed to news, causing domain mismatch.

These issues motivate fine-tuning, domain adaptation, and human oversight. Targeted adaptation–finetuning on modest in-domain data–yields substantial metric improvements (BLEU, chrF). Domain shifts (general to legal/medical) significantly degrade quality without adaptation or guided constraints.

2.3 Commercial vs. Academic & Open Approaches

Commercial systems (Google Translate, Microsoft Translator, DeepL) are strong baselines for Arabic, offering reliable MSA performance in general domains, with features for professional workflows (document translation, glossary management, enterprise integration).

Modern LLM-based translation, especially in interactive/constrained settings, is highly competitive, producing fluent output adaptable via prompting for tone/terminology. Commercial engines remain preferable for predictability, low latency, deterministic behavior, or compliance-sensitive production services.

Academic & open-source approaches include multilingual MT models and open LLMs fine-tunable for Arabic. Large multilingual translation models and open LLMs with strong Arabic coverage enable customization (LoRA/QLoRA fine-tuning, translation memory integration, local deployment) but require engineering to match commercial usability/robustness.

Evaluation remains central. BLEU is common but can under-represent fluency/adequacy improvements for morphologically rich languages like Arabic. Character-level measures (chrF) better accommodate inflectional/orthographic variation. High-quality Arabic evaluation combines automatic metrics with human assessment for professional standards of readability, cultural sensitivity, and domain correctness.

2.4 Human-in-the-Loop Translation Environments

HITL systems combine MT speed/coverage with human judgment/cultural competence, crucial for Arabic nuance, register, and appropriateness. Traditional computer-assisted translation (CAT) tools use translation memories (TM), terminology databases, and MT suggestions. The shift is toward **LLM-assisted HITL environments**, where LLMs act as collaborative assistants.

Key research and development directions:

1. **Interactive translation & iterative refinement:** Systems support feedback loops where translator corrections (terminology, tone) update outputs via prompting, without retraining.
2. **Multi-candidate generation & comparison:** Generating multiple candidates (from multiple models/decoding strategies) for side-by-side presentation

lets translators select, blend, and edit efficiently, reducing rewrite time and offering stylistic options.

3. **Terminology control & translation memories:** Integrating TM/glossary constraints directly into generation ensures consistent term use; LLMs can rephrase while enforcing terms or flag mismatches.
4. **Document ingestion, segmentation, & workflow automation:** Robust parsing of documents (PDF/Word), segmentation into meaningful units, and polished re-export are essential for usability, though non-trivial for complex layouts.
5. **Safety, bias, & trustworthiness:** Keeping humans responsible for final decisions, with safeguards (filtering unsafe outputs, flagging hallucinations, showing suggestion provenance) mitigates risks of subtle errors or culturally inappropriate phrasing.

HITL is critical for domains requiring human review (legal, medical, government, public communications) for ethical/regulatory reasons; MT accelerates drafting but doesn't replace expert oversight.

2.5 Implications for the Proposed System

The proposed system builds an LLM-assisted environment for English/French-Arabic translation integrating: (i) curated parallel corpora/TMs, (ii) evaluation/fine-tuning of open-source models, and (iii) a translator interface for document ingestion, segmentation, multi-model generation, interactive editing, and export.

State-of-the-art conclusions:

- Arabic quality improves substantially via data curation, domain adaptation, and terminology control, especially with parameter-efficient fine-tuning (LoRA/QLoRA).
- Commercial systems are strong baselines, but LLM-based translation is competitive when translators guide output via prompts and controlled terminology.
- The most promising approach is hybrid workflows with human control, using LLMs for strong drafts, alternatives, and editing assistance.
- Evaluation should combine automatic metrics (BLEU, chrF, semantic measures) with structured human review, including interface usability and post-editing effort measurement.

Overall, the highest-impact systems for Arabic treat translation as a human-centered workflow supported by LLMs—prioritizing consistency, controllability, and cultural/domain appropriateness—over optimizing one-shot automatic MT scores.

3 Data Set Building

3.1 Overview

The data collection phase focused on assembling a comprehensive multilingual parallel corpus spanning five specialized domains: education, general content, medical, legal, and journalistic texts. The final dataset comprises approximately 6GB of data in JSONL format, covering Arabic, English, and French language pairs. All sources were selected from open-access repositories with appropriate licensing for academic research.

3.2 Domain-Specific Data Collection

3.2.1 Educational Domain

Educational data was primarily sourced from the OPUS repository. The **TED2020 Corpus** contains professionally translated TED talk transcripts covering informal and semi-formal educational topics, with 350,000 sentence pairs each for English $\leftrightarrow$ Arabic and French $\leftrightarrow$ Arabic (700,000 total). Additional OPUS sources including QED, NeuLab TED Talks, and GNOME corpora supplemented the educational collection. Web scraping was conducted exclusively on Wikibooks and Wikiversity, selected for their openly licensed educational materials under Creative Commons licenses.

3.2.2 General Domain

The United Nations Parallel Corpus (v1.0) served as the primary source for general-domain data, consisting of manually translated official UN documents. From an original size of approximately 7.5GB per language, 3.5 million sentence pairs were selected for each language pair (EN-AR and FR-AR). The corpus uses a line-aligned format where corresponding sentences across languages share identical line numbers, eliminating the need for additional alignment algorithms.

3.2.3 Medical Domain

Medical data was collected from multiple specialized healthcare corpora: **ELRC3083 Wikipedia Health (EN-AR)**, **ELRC Wikipedia Health (FR-AR)**, a **Human-Evaluated Medical Dataset** (retaining only highest-quality pairs scored 4/4 by professionals), and the **PEACH Dataset** (51,671 sentence pairs from patient information leaflets).

3.2.4 Legal Domain

Legal domain data was extracted from the UN Parallel Corpus v1.0, consisting of official records and parliamentary documents in the public domain. This corpus contains

86,307 aligned documents and 11,365,709 aligned sentence pairs, professionally translated between 1990 and 2014. Content includes international treaties, UN resolutions, legal reports, and human rights documentation.

3.2.5 Journalistic Domain

The News-Commentary dataset from OPUS was selected as the primary source, providing professionally curated news and opinion pieces with consistent sentence-level alignment. An initial attempt to manually collect parallel news data was abandoned due to the impracticality of reliable sentence-level alignment from independent reporting.

3.3 Data Standardization

All collected data across all five domains was converted into a standardized JSONL format to ensure consistency. Each entry follows a uniform schema specifying `source_language`, `target_language`, `source_text`, and `target_text`. This structure facilitates efficient data processing, model training, and streaming operations.

3.4 Dataset Characteristics and Statistics

The final dataset totals approximately 6GB. Approximate distribution:

- Educational: 700,000+ pairs
- General: 7,000,000 pairs (3.5M per language pair)
- Medical: 100,000+ pairs
- Legal: 11,365,709 aligned pairs
- Journalistic: 300,000+ pairs

The dataset comprehensively covers three language pairs with bidirectional translation capabilities: English↔Arabic, French↔Arabic, and implicitly English↔French.

3.5 Quality Assurance

Multiple quality control measures were implemented: selection of sources from reputable repositories (OPUS, UN, academic datasets); incorporation of human-evaluated translations in the medical domain; verification of license compliance for academic research; and alignment verification during preprocessing to filter out misaligned or corrupted pairs.

3.6 Linguistic Characteristics

The corpus exhibits diverse linguistic characteristics: register variation (from informal TED talks to formal UN documents); specialized domain terminology; varying sentence length and complexity; temporal coverage (e.g., 1990-2014 in UN corpus); and culturally specific content requiring careful handling.

3.7 Arabic Preprocessing

Arabic text required extensive preprocessing due to its complex orthographic variations and morphological structure.

3.7.1 Normalization

Multiple character variants were normalized to their standard forms. Alif variants (ا, آ, إ, ٱ) were normalized to the base form (ا). Ya variants including Alif maqsura (ي), Persian ya (ی), and ya with hamza (ئ) were normalized to standard ya (ي). Arabic punctuation marks were converted to their English equivalents: Arabic comma (،) to comma (,), Arabic semicolon (؛) to semicolon (;), Arabic question mark (؟) to question mark (?), and Arabic quotation marks (ﻋﺒﺮﺓ) to standard quotes (""). Eastern Arabic numerals (٠-٩) were converted to Western Arabic numerals (0-9).

3.7.2 Character Removal

Emojis, Quranic stop signs, ornamental marks, unnecessary Latin characters, diacritics, and extra spaces were removed. Spacing around punctuation was fixed and newlines were normalized.

3.7.3 Orthographic Normalization

Various quote styles («النص», "النص", "النص", "النص") were unified to standard ASCII quotes. Multiple dash characters (hyphen, en dash, em dash, Arabic tatweel, minus sign, underscore, non-breaking hyphen) were normalized to simple ASCII hyphens. Unicode normalization (NFC/NFKC) was applied to ensure consistent character encoding, converting presentation forms and removing invisible characters like zero-width joiners to prevent tokenization issues.

3.8 English and French Preprocessing

For English and French text, standard preprocessing steps included lowercasing to reduce vocabulary size and normalize text representation. Punctuation and whitespace normalization was performed to remove HTML artifacts and fix inconsistent spacing.

Language-specific rules were applied to French text, treating apostrophes and liaisons carefully to preserve linguistic integrity. Diacritics in French were typically retained unless the downstream model specifically benefited from their removal, as they carry important semantic information.

4 System Design

4.1 Pre-Fine-Tuning Evaluation

A baseline evaluation was conducted to assess the translation capabilities of candidate multilingual models prior to fine tuning. The evaluation focused on bidirectional English-Arabic (EN-AR) and French-Arabic (FR-AR) translation. More details are found in the next section.

4.2 Model Selection

Based on the comprehensive evaluation across 28,000 translation pairs ($7,000 \times 4$ directions), three models were selected:

- **Facebook NLLB-200-1.3B**: Best balance of performance and efficiency
- **Facebook M2M100-1.2B**: Strong bidirectional performance, well-optimized
- **Facebook NLLB-200-3.3B**: Highest quality for production-critical applications

Selection criteria included translation quality across the necessary directions (FR - AR and EN - AR), inference speed, memory requirements, and deployment feasibility.

4.3 Fine Tuning Configuration

Table 1 summarizes the key differences between the two fine-tuning configurations:

Table 1: Comparison of Fine-Tuning Configurations

Parameter	NLLB-3.3B	NLLB-1.3B
Model Parameters	3.3B	1.3B
LoRA Rank	32	16
LoRA Alpha	64	32
Learning Rate	4×10^{-4}	5×10^{-4}
Batch Size	4	16
Effective Batch Size	32	64
Training Epochs	3	3
Data Subset Ratio	20%	35%
Target Modules	6 layers	4 layers

Both configurations employed QLoRA methodology, enabling efficient adaptation of large multilingual models while maintaining high translation quality.

4.4 Human-in-the-Loop Agent Architecture

The system implements a multi-agent pipeline for collecting, validating, deduplicating, and managing human translation corrections. This ensures continuous learning while maintaining strict quality control.

4.4.1 Agent Components and Responsibilities

1. Preprocessor Agent *Normalizes and structures incoming human edits.*

- Arabic normalization: Removes diacritics, standardizes letters, converts Arabic digits
- Latin scripts: Standardizes quotes, apostrophes, hyphens
- Language detection and validation
- Returns structured JSON with normalized source, LLM outputs, and human edit

2. Validator Agent *Evaluates human corrections for quality improvements.*

Decision Flow:

1. **Exact Duplication:** Embedding similarity $> 0.995 \rightarrow$ REJECT
2. **Semantic Verification:** LLM checks meaning preservation $\rightarrow$ REJECT if meaning altered
3. **Near-Duplication (≥ 0.92):** LLM judges if edits provide clear improvement
4. **Quality Improvement (< 0.92):** LLM determines if human is clearly better
5. **Confidence Scoring:** Weighted score (CHRF: 40%, length: 10%, LLM: 50%)
6. **Acceptance:** Score $\geq 0.70 \rightarrow$ ACCEPT

Implementation Details:

```
class ValidatorAgent:
    def __init__(self, accept_threshold=0.70,
                  exact_threshold=0.995, near_threshold=0.92):
        self.embedder = SentenceTransformer(
            "paraphrase-multilingual-MiniLM-L12-v2"
        )
        self.judge_model = AutoModelForCausalLM.from_pretrained(
            "Qwen/Qwen2.5-0.5B-Instruct"
        )
```

```
def validate(self, source, llm_outputs, human):  
    # Compute similarity scores  
    # Run semantic verification  
    # Judge quality improvement  
    # Compute final score  
    # Return decision with reasoning
```

3. Deduplicator Agent *Ensures uniqueness using FAISS vector similarity search.*

- Maintains FAISS index of all accepted sentences
- Sentence segmentation for granular duplicate detection
- Similarity threshold: 0.85 (reject if any sentence exceeds)
- Preloads from existing `master.jsonl`
- Incremental updates with new acceptances

4. Executor Agent *Persists validated corrections to disk.*

- JSON schema validation before saving
- Newline-delimited JSON format (`master.jsonl`)
- Only saves "accepted" decisions
- Clean storage (removes internal fields)

5. Monitor Agent *Manages fine-tuning cycles and data accumulation.*

Triggers:

- **Count-based:** 10,000 corrections in `master.jsonl`
- **Time-based:** 7 days since last reset

Functions:

- Prepares dataset for fine-tuning
- Builds few-shot prompts from last 50 corrections
- Archives current data and resets system
- Clears FAISS memory for new cycle

6. Orchestrator Agent *Coordinates the entire pipeline.*

- Batch processing interface for frontend
- Sequences agents: Preprocessor → Validator → Deduplicator → Executor
- Error handling and result aggregation
- Automatic monitor checking after processing

4.4.2 Data Flow Diagram

Frontend → Preprocessor → Validator → Deduplicator → Executor → Monitor

Model Loading Strategy:

- Embedding model: Multilingual MiniLM-L12-v2 (cached locally)
- Judge model: Qwen2.5-0.5B-Instruct (supports .safetensors/.bin)
- Device management: Automatic CUDA with CPU fallback

4.5 Integration and Workflow

4.5.1 Model Integration

- Selected models (NLLB-1.3B, M2M100-1.2B, NLLB-3.3B) connected to frontend/backend
- Human corrections flow through agentic pipeline
- Validated entries stored in `master.jsonl`

4.5.2 Continuous Learning Cycle

1. Human corrects model translations via frontend
2. Correction processed through agent pipeline
3. Accepted corrections accumulate in `master.jsonl`
4. Monitor triggers fine-tuning at thresholds
5. New model versions improve translation quality
6. Cycle repeats with improved models

5 Results and Analysis

5.1 Evaluation Metrics

The translation models are evaluated using widely adopted automatic machine translation metrics: **BLEU** (n-gram overlap), **METEOR** (semantic similarity), **TER** (Translation Edit Rate; lower is better), **chrF++** (character-level metric), and **COMET** (neural-based metric for meaning preservation).

5.2 Initial Model Evaluation and Selection

Before fine-tuning, different models were evaluated across four language pairs on a sample of 7000 examples. The comprehensive results (average scores across all directions) are presented in Table 2.

Table 2: Initial Evaluation of Translation Models (Average Scores Across All Directions)

Model	BLEU	METEOR	TER	chrF++	COMET
NLLB-200-600M	2.73	7.61	113.94	8.00	51.82
MBART-50 (2.1B)	2.00	5.16	118.03	3.41	46.35
Qwen2-1.5B	0.42	6.35	561.61	7.43	38.42
Qwen-3-4B	8.56	26.33	77.67	37.48	64.45
M2M-100	—	28.05	81.45	42.47	—
Gemma-2-2B	9.76	28.59	85.54	36.78	70.83
MADLAD-3.3B	27.54	45.09	63.42	57.82	81.76
NLLB-3.3B	21.24	41.45	69.01	48.58	80.29
NLLB-1.3B	—	40.30	70.85	51.20	76.60
Helsinki-NLP	40.22	67.10	42.98	—	—

5.2.1 Model Selection

Based on evaluation metrics, inference time, and computational efficiency, three models were selected for fine-tuning:

1. **NLLB-200 (1.3B)**: Strong balance between performance and efficiency.
2. **NLLB-3.3B**: Superior translation performance and robust multilingual capabilities.
3. **M2M-100 (1.2B)**: Balanced multilingual performance and manageable computational requirements.

Note: Due to time constraints during M2M-100 fine-tuning, the Helsinki-NLP model was temporarily used for demonstration, providing comparable baseline performance.

5.3 Fine-Tuned Model Performance

5.3.1 Model 1: NLLB-200 (1.3B)

After fine-tuning, the NLLB-200 model demonstrated significant improvements across all metrics, with results shown in Table 3. The excellent COMET score reflects strong semantic adequacy and meaning preservation.

Table 3: NLLB-200 (1.3B) After Fine-Tuning

Metric	Average Score	Quality
BLEU	30.27	Acceptable
METEOR	59.49	Good
TER	48.98	Acceptable
chrF++	66.89	Good
COMET	91.66	Excellent

5.3.2 Model 2: NLLB-3.3B

Fine-tuning led to substantial improvement for NLLB-3.3B, with notable reduction in TER and strong gains in semantic similarity. The results, shown in Table 4, represent the best overall performance achieved.

Table 4: NLLB-3.3B After Fine-Tuning

Metric	Average Score	Quality
BLEU	48.81	Good
METEOR	73.24	Very Good
TER	30.52	Good
chrF++	76.57	Very Good
COMET	—	—

5.3.3 Model 3: Helsinki-NLP (Baseline)

The Helsinki-NLP model, evaluated without fine-tuning, achieves competitive baseline performance (Table 5) due to extensive OPUS pretraining, though it remains less adaptable to domain-specific content.

5.4 Fine-Tuning Impact Comparison

The results clearly show that fine-tuning leads to substantial improvements in translation quality across all evaluated metrics. Both NLLB-200 (1.3B) and NLLB-3.3B models benefited significantly from domain-specific fine-tuning, with consistent gains

Table 5: Helsinki-NLP Without Fine-Tuning

Metric	Average Score	Quality
BLEU	40.22	Moderate
METEOR	67.10	High
TER	42.98	Moderate

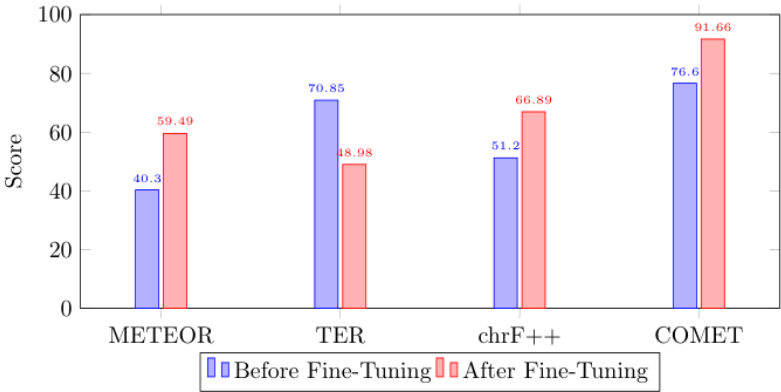


Figure 1: NLLB-200 (1.3B): Performance Before and After Fine-Tuning

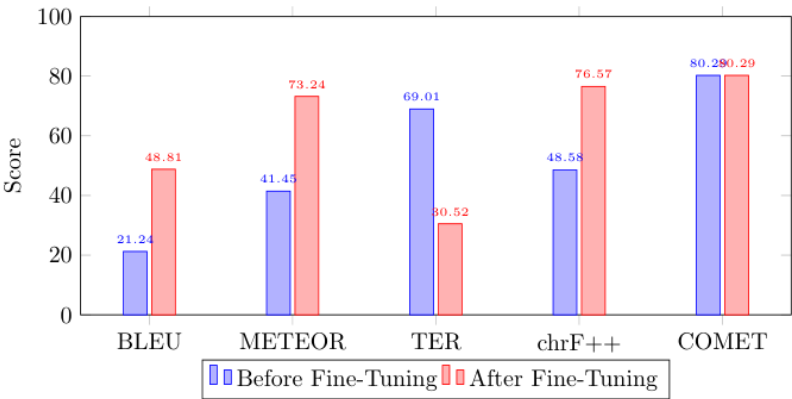


Figure 2: NLLB-3.3B: Performance Before and After Fine-Tuning

in METEOR, chrF++, and COMET, alongside notable reductions in TER. The NLLB-3.3B model achieved the most pronounced improvements, particularly in BLEU and TER, highlighting its superior capacity to learn complex linguistic patterns.

6 Discussion

The experimental results demonstrate that fine-tuning large multilingual models significantly improves Arabic machine translation quality. Using a combination of automatic metrics (BLEU, METEOR, TER, chrF++, and COMET), the analysis shows clear gains in both lexical accuracy and semantic adequacy after domain-specific training.

Among the evaluated models, **NLLB-3.3B achieved the best overall performance after fine-tuning**. It recorded the highest BLEU and METEOR scores, alongside a substantial reduction in TER, indicating fewer post-editing operations would be required. The strong chrF++ results further suggest improved handling of Arabic morphology, which is particularly important given the language’s rich inflectional structure. These results highlight the model’s ability to capture complex linguistic patterns when sufficient model capacity and training data are available.

The **NLLB-200 (1.3B) model also showed notable improvements** after fine-tuning. Although its absolute performance remained slightly below that of NLLB-3.3B, it achieved an excellent COMET score, reflecting strong meaning preservation. This indicates that the model produces translations that are semantically faithful to the source, even if surface-level variations remain. Importantly, NLLB-200 offers a better balance between translation quality and computational efficiency, making it more suitable for deployment in resource-constrained environments.

The **Helsinki-NLP model**, evaluated without fine-tuning, delivered competitive baseline results, especially in METEOR and BLEU. This performance can be attributed to extensive pretraining on large-scale OPUS parallel corpora. However, its lack of fine-tuning limited its adaptability to domain-specific and paragraph-level translations, confirming that pretraining alone is insufficient for optimal performance in specialized contexts.

Despite these promising results, several **limitations** should be noted. First, evaluation relied exclusively on automatic metrics, which, although widely used, do not fully capture translation fluency, cultural appropriateness, or contextual nuance. Second, time and computational constraints limited the extent of fine-tuning, particularly for the M2M-100 model, which prevented a complete comparative analysis. Additionally, the dataset size, while sufficient to demonstrate improvements, may not cover all linguistic variations present in real-world Arabic translation scenarios.

Future work includes adding human evaluation, expanding dialectal and domain diversity, using human in the loop feedback, and optimizing fine tuning for lower computational cost. Overall, the findings show that domain specific fine tuning significantly improves Arabic machine translation, with larger adapted models achieving better translation quality.

7 Conclusion

This project successfully developed a human-in-the-loop (HITL) framework for English/French–Arabic machine translation, addressing Arabic’s linguistic complexity by integrating automated systems with human expertise. Our contributions include creating a specialized parallel corpus across five domains, demonstrating through systematic evaluation that parameter-efficient fine-tuning of the NLLB-3.3B model yields superior performance, and designing an innovative multi-agent pipeline that enables continuous learning from human corrections. Key findings confirm that domain-specific adaptation is essential for specialized contexts, and that while automatic metrics provide valuable benchmarks, human evaluation remains crucial for assessing fluency and cultural nuance. The proposed system offers a sustainable, human-centered model for improving translation quality, serving as a reference framework for future collaborative translation systems.

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8 Appendix A

#1 Paragraph

Accepted

SOURCE

We are a group of fourth-year students working on a human-in-the-loop machine translation system that combines neural models with real human feedback. Our goal is to build a platform where English texts are automatically translated into Arabic by several large translation models, but the final decision is always guided and improved by human reviewers. In our system, documents are loaded and divided into smaller parts, then passed through multiple models like NLLB and M2M to generate diverse candidate translations. These model outputs are displayed in a web interface where human annotators can compare them, edit them, or write an entirely new translation if necessary. Then each human correction is analyzed and stored so it can be used later to improve and continuously enhance model quality.

MODEL OUTPUTS

Model 1

Model 2

Model 3

نحن مجموعة من طلاب السنة الرابعة نعمل على نظام ترجمة آلي بشري في حلقة يجمع بين النماذج العصبية مع ردود الفعل البشرية الحقيقية. هدفنا هو بناء منصة حيث يتم ترجمة النصوص الإنجليزية تلقائياً إلى العربية من قبل العديد من نماذج الترجمة الكبيرة، ولكن يتم توجيه القرار النهائي دائماً وتحسينه من قبل المراجعين البشريين. في نظامنا، يتم تحميل المستندات وتقسيمها إلى أجزاء أصغر، ثم تمريرها من خلال نماذج متعددة مثل NLLB و M2M لتوليد ترجمات مرشحة متنوعة. يتم عرض نتائج النموذج هذه في واجهة ويب حيث يمكن للمعلق البشري مقارنتها وتحريرها أو كتابة ترجمة جديدة تماماً إذا لزم الأمر. ثم يتم تحليل كل تصحيح بشري وتحليله وتخزينه بحيث يمكن استخدامه لاحقاً لتحسين النماذج وتحسين جودتها باستمرار.

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نحن مجموعة من طلاب السنة الرابعة نعمل على نظام ترجمة آلي بشري في حلقة يجمع بين النماذج العصبية مع ردود الفعل البشرية الحقيقية.

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Figure 3: Translation English → Arabic Model 1

#1 Paragraph

Accepted

SOURCE

We are a group of fourth-year students working on a human-in-the-loop machine translation system that combines neural models with real human feedback. Our goal is to build a platform where English texts are automatically translated into Arabic by several large translation models, but the final decision is always guided and improved by human reviewers. In our system, documents are loaded and divided into smaller parts, then passed through multiple models like NLLB and M2M to generate diverse candidate translations. These model outputs are displayed in a web interface where human annotators can compare them, edit them, or write an entirely new translation if necessary. Then each human correction is analyzed and stored so it can be used later to improve and continuously enhance model quality.

MODEL OUTPUTS

Model 1

Model 2

Model 3

نحن مجموعة من طلاب السنة الرابعة الذين يعملون على نظام ترجمة آلية من الإنسان في الحلقة يجمع بين النماذج العصبية مع ردود الفعل البشرية الحقيقية. هدفنا هو بناء منصة يتم فيها ترجمة النصوص الإنجليزية تلقائياً إلى العربية بواسطة عدة نماذج ترجمة كبيرة، ولكن القرار النهائي دائماً ما يتم توجيهه وتحسينه بواسطة المراجعين البشريين. في نظامنا، يتم تحميل الوثائق وتقسيمها إلى أجزاء أصغر، ثم تمرير نماذج متعددة مثل NLLB و M2M لتوليد ترجمات مرشحة متنوعة. يتم عرض مخرجات النموذج في واجهة ويب حيث يمكن للمعلق البشري مقارنة بها أو تحريرها أو كتابة ترجمة جديدة تماماً إذا لزم الأمر. ثم يتم تحليل كل تصحيح بشري ثم يتم التحقق من صحتها وتخزينها بحيث يمكن استخدامها لاحقاً لتنظيم النماذج وتحسين جودتها.

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نحن مجموعة من طلاب السنة الرابعة نعمل على نظام ترجمة آلي بشري في حلقة يجمع بين النماذج العصبية مع ردود الفعل البشرية الحقيقية.

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Figure 4: Translation English → Arabic Model 2

#1 Paragraph

Accepted

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We are a group of fourth-year students working on a human-in-the-loop machine translation system that combines neural models with real human feedback. Our goal is to build a platform where English texts are automatically translated into Arabic by several large translation models, but the final decision is always guided and improved by human reviewers. In our system, documents are loaded and divided into smaller parts, then passed through multiple models like NLLB and M2M to generate diverse candidate translations. These model outputs are displayed in a web interface where human annotators can compare them, edit them, or write an entirely new translation if necessary. Then each human correction is analyzed and stored so it can be used later to improve and continuously enhance model quality.

MODEL OUTPUTS

Model 1

Model 2

Model 3

نحن مجموعة من طلاب السنة الرابعة يعملون على نظام ترجمة آلي بشرية تتكامل مع النماذج العصبية مع ردود الفعل البشرية الحقيقية. هدفنا هو بناء منصة حيث يتم ترجمة النصوص الإنجليزية بشكل تلقائي إلى اللغة العربية من خلال عدة نماذج ترجمة كبيرة، ولكن القرار النهائي يتم القيام به دائمًا من قبل المراجعين البشرية وتحسينه. في نظامنا، يتم تحميل الوثائق، وتقسيمها إلى أجزاء صغيرة، ثم يتم نقلها من خلال العديد من النماذج مثل NLLB و M2M لإنتاج ترجمة مرشحة متنوعة. يتم عرض هذه النماذج في رابط ويب حيث يمكن للمرشح البشري مقارنة تلك النماذج، إعدادها أو كتابة ترجمة جديدة تمامًا إذا لزم الأمر.

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نحن مجموعة من طلاب السنة الرابعة نعمل على نظام ترجمة آلي بشري في حلقة يجمع بين النماذج العصبية مع ردود الفعل البشرية الحقيقية.

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Figure 5: Translation English → Arabic Model 3

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Le développement rapide de l'intelligence artificielle soulève des enjeux majeurs pour les institutions académiques, tant sur le plan méthodologique qu'éthique. Dans le domaine de la recherche, ces technologies offrent de nouveaux outils d'analyse de données massives, permettant d'identifier des tendances complexes qui seraient difficiles à détecter avec des méthodes classiques. Parallèlement, elles obligent les chercheurs à repenser la transparence des modèles, la reproductibilité des expériences et la protection des données sensibles utilisées pour l'entraînement. Les universités doivent donc adapter leurs programmes de formation afin de doter les étudiants de compétences solides en statistiques, en programmation et en réflexion critique sur les biais algorithmiques. Cette évolution vise à garantir que l'IA soit intégrée de manière responsable, en respectant les principes de rigueur scientifique et les valeurs fondamentales de la recherche académique.

MODEL OUTPUTS

Model 1

Model 2

Model 3

تير التطور السريع للذكاء الاصطناعي تحديات كبيرة للمؤسسات الأكاديمية، من الناحية المنهجية والأخلاقية. في مجال البحث، تقدم هذه التقنيات أدوات جديدة لتحليل البيانات الضخمة، مما يسمح بتحديد الاتجاهات المعقدة التي سيكون من الصعب اكتشافها باستخدام الأساليب التقليدية. في الوقت نفسه، فإنها تفرض على الباحثين إعادة التفكير في شفافية النماذج، وإمكانية إعادة التجربة، وحماية البيانات الحساسة المستخدمة للتدريب. وبالتالي، يجب على الجامعات تكيف برامج التدريب الخاصة بها لتزويد الطلاب بمهارات قوية في الإحصاء والبرمجة والتفكير النقدي في التحيز الخوارزمي. يهدف هذا التطور إلى ضمان إدماج الذكاء الاصطناعي بطريقة مسؤولة، مع احترام مبادئ الصرامة العلمية والقيم الأساسية للبحث الأكاديمي.

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Figure 6: Translation French → Arabic Model 1

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Le développement rapide de l'intelligence artificielle soulève des enjeux majeurs pour les institutions académiques, tant sur le plan méthodologique qu'éthique. Dans le domaine de la recherche, ces technologies offrent de nouveaux outils d'analyse de données massives, permettant d'identifier des tendances complexes qui seraient difficiles à détecter avec des méthodes classiques. Parallèlement, elles obligent les chercheurs à repenser la transparence des modèles, la reproductibilité des expériences et la protection des données sensibles utilisées pour l'entraînement. Les universités doivent donc adapter leurs programmes de formation afin de doter les étudiants de compétences solides en statistiques, en programmation et en réflexion critique sur les biais algorithmiques. Cette évolution vise à garantir que l'IA soit intégrée de manière responsable, en respectant les principes de rigueur scientifique et les valeurs fondamentales de la recherche académique.

MODEL OUTPUTS

Model 1

Model 2

Model 3

يثير التطور السريع لذكاء اصطناعي تحديات كبيرة للمؤسسات الأكاديمية، سواء من الناحية المنهجية أو الأخلاقية. في مجال البحث، توفر هذه التقنيات أدوات جديدة لتحليل البيانات الضخمة، مما يسمح بتحديد اتجاهات معقدة من الصعب اكتشافها بالطرق التقليدية. في الوقت نفسه، فإنها تجبر الباحثين على إعادة التفكير في شفافية النماذج، وتكرار التجارب، وحماية البيانات الحساسة المستخدمة في التدريب. لذلك يجب على الجامعات تغيير برامجها التدريبية لتزويد الطلاب بمهارات قوية في الإحصاءات، والبرمجة، والتفكير النقدي في التحيزات الخوارزمية. تهدف هذه التطورات إلى ضمان أن تكون الـ IA مدمجة بشكل مسؤول، وذلك باحترام المبادئ الأساسية للقوة العلمية والقيم الصارمة للبحث الأكاديمي.

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Figure 7: Translation French → Arabic Model 2

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Le développement rapide de l'intelligence artificielle soulève des enjeux majeurs pour les institutions académiques, tant sur le plan méthodologique qu'éthique. Dans le domaine de la recherche, ces technologies offrent de nouveaux outils d'analyse de données massives, permettant d'identifier des tendances complexes qui seraient difficiles à détecter avec des méthodes classiques. Parallèlement, elles obligent les chercheurs à repenser la transparence des modèles, la reproductibilité des expériences et la protection des données sensibles utilisées pour l'entraînement. Les universités doivent donc adapter leurs programmes de formation afin de doter les étudiants de compétences solides en statistiques, en programmation et en réflexion critique sur les biais algorithmiques. Cette évolution vise à garantir que l'IA soit intégrée de manière responsable, en respectant les principes de rigueur scientifique et les valeurs fondamentales de la recherche académique.

MODEL OUTPUTS

Model 1

Model 2

Model 3

إن تطوير الذكاء الاصطناعي سريع يهدد المؤسسات الأكاديمية بحدوث مشكلات كبيرة، سواء من حيث المنهجية أو الأخلاقية. في مجال البحوث، تقدم هذه التقنيات أدوات تحليلية جديدة للبيانات الكبيرة، مما يتيح تحديد الاتجاهات المعقدة التي من الصعب اكتشافها باستخدام الأساليب الكلاسيكية. في الوقت نفسه، يطلب من الباحثين التفكير في إعادة التفكير في الشفافية من النماذج، والتكرار من التجربة، وحماية البيانات الحساسة المستخدمة للتدريب. لذلك، يجب على الجامعات تحديد برامج تدريبهم لتزويد الطلاب بالمهارات القوية في الإحصائيات، والتنظيم، والتفكير في التفكير النقدي حول التغيرات الأكاديمية.

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إن تطوير الذكاء الاصطناعي سريع يهدد المؤسسات الأكاديمية بحدوث مشكلات كبيرة، سواء من حيث المنهجية أو الأخلاقية. في مجال البحوث، تقدم هذه التقنيات أدوات تحليلية جديدة للبيانات الكبيرة، مما يتيح تحديد الاتجاهات المعقدة التي من الصعب اكتشافها باستخدام الأساليب الكلاسيكية. في الوقت نفسه، يطلب من الباحثين التفكير في إعادة التفكير في الشفافية من النماذج، والتكرار من التجربة، وحماية البيانات الحساسة المستخدمة للتدريب. لذلك، يجب على الجامعات تحديد

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Using model3 as base

Figure 8: Translation French → Arabic Model 3

9 Appendix B: Who did what

Member	Contributions
Lina-Wafaa BENTIBA	• Collection and pre-processing of the legal domain data • Evaluation of the translation on the model NLLB-200-3.3B • Fine tuning NLLB-200-3.3B • Contributed to report writing
Feriel ROUAINIA	• Collection and pre-processing of the general domain data • Evaluation on facebook/nllb-200-distilled-600M, MBART-50 Large, Qwen2-1.5B • Fine tuning NLLB-200 (1.3B) • Contributed to report writing
Sara ABID	• Collection and pre-processing of the education domain data • Evaluation on Qwen-3-4B and Gemma-2-2B • HITL architecture design and implementation of three agents • Contributed to report writing
Imane AISSAT	• Collection and pre-processing of the medical domain data • Evaluation on MADLAD 3.3B and LMT 600 1.7B • Implementation of three agents and HITL pipeline integration • Organization of the final notebook code
Younes AHMANE	• Development of the frontend and backend of the platform • Contributed to report writing
Takoua HIDOUSI	• Collection and pre-processing of the journalistic domain data • Evaluation on M2M-100-1.2B and NLLB-1.3B • Initial fine tuning of M2M-100-1.2B • Contributed to report writing